Aalto University
Department of Mathematics and Systems Analysis
MS-C1300 — Complex Analysis, 2026-2027/II

Problem set 3B

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Exercise sessions: Thu-Fri, Nov 5-6, 2026

Topic: contour integrals, primitives, Cauchy's integral theorem

The first two exercises (marked with symbol ♮) are to be solved before the exercise sessions so that you are ready to present your solution on the blackboard. The last two exercises are to be discussed and solved in exercise sessions.

♮ Blackboard exercise 1.
Let $\gamma\colon [0,2\pi] \to \mathbb{C}$ be the path given by the formula $\gamma(t) = e^{\mathrm{i}t}$. Evaluate the following integrals along γ:

(a) $\displaystyle\int_\gamma \frac{1}{(z-2)^2}\,\mathrm{d}z$

(b) $\displaystyle\int_\gamma \frac{1}{z^2-4}\,\mathrm{d}z$

(c) $\displaystyle\int_\gamma \Big(z+\frac{1}{z}\Big)^n\,\mathrm{d}z$ where $n \in \mathbb{N}$.

♮ Blackboard exercise 2.
Consider the closed contours γ_Δ and σ defined as follows. Let γ_Δ be the closed contour that traverses the boundary of the triangle with vertices 0, 1 and i once counterclockwise, i.e., the concatenation of the line segments $[0,1]$, $[1,\mathrm{i}]$ and $[\mathrm{i},0]$. To define σ, fix $c \in \mathbb{C}$ and $r > 0$, and let $\sigma(t) = c + re^{\mathrm{i}t}$ for $t \in [0,2\pi]$.

(a) Calculate $\displaystyle\oint_{\gamma_\Delta} \overline{z}\,\mathrm{d}z$.

(b) Calculate $\displaystyle\oint_{\sigma} \overline{z}\,\mathrm{d}z$.

(c) Both (a) and (b) yielded $2\mathrm{i}$ times the area enclosed by the contour. Use this, the linearity of the integral, and the fact that $z \mapsto z$ has a primitive, to calculate

$$\oint \mathfrak{Re}(z)\,\mathrm{d}z \quad \text{and} \quad \oint \mathfrak{Im}(z)\,\mathrm{d}z$$

along γ_Δ and along σ. Do the functions $z \mapsto \mathfrak{Re}(z)$ and $z \mapsto \mathfrak{Im}(z)$ have primitives on $\mathbb{C}$?

Hint: *$\mathfrak{Re}(z)$ and $\mathfrak{Im}(z)$ can be written in terms of z and $\bar{z}$.*

Exercise (in class) 3.
Let $a \in \mathbb{R}$ be a real number such that $|a| > 1$. With the help of the complex integral

$$\int_\gamma \frac{1}{z+a}\,\mathrm{d}z$$

along the path $\gamma\colon [0, 2\pi] \to \mathbb{C}$ given by the formula $\gamma(t) = e^{\mathrm{i}t}$, show that

$$\int_0^{2\pi} \frac{1 + a\,\cos(t)}{1 + 2a\,\cos(t) + a^2}\,\mathrm{d}t \;=\; 0.$$

Exercise (in class) 4.
Consider the following open subsets of $\mathbb{C}$:

$$\begin{aligned}
A &= \left\{z \in \mathbb{C} \;\middle|\; |z+1| < \tfrac{3}{2}\right\} \cup \left\{z \in \mathbb{C} \;\middle|\; |z-1| < \tfrac{3}{2}\right\},\\
B &= \left\{z \in \mathbb{C} \;\middle|\; \mathfrak{Re}(z) > 0\right\} \cup \left\{z \in \mathbb{C} \;\middle|\; \mathfrak{Im}(z) > 0\right\},\\
C &= \left\{z \in \mathbb{C} \;\middle|\; |z| < 1\right\} \setminus \left\{\tfrac{1}{2}\right\},\\
D &= \left\{z \in \mathbb{C} \setminus \{0\} \;\middle|\; |\mathrm{Arg}(z)| < \tfrac{\pi}{3}\right\}.
\end{aligned}$$

(a) Which of these sets are convex? Which are star-shaped?
(b) For one of the sets above, call it X, find an analytic $f\colon X \to \mathbb{C}$ and a closed contour γ in X for which $\oint_\gamma f(z)\,\mathrm{d}z \neq 0$.
Hint: *Part (a) and Cauchy's integral theorem may guide you to which of the sets you may choose as X in (b).*